InternHub - engineering requirements

1. Introduction and Purpose

a. Project Overview

InternHub is a platform designed to manage internship offers and student applications. The system allows students to discover internship opportunities, apply for suitable positions, track the status of their applications, and withdraw applications when the business rules allow it. The platform also supports the management of internship offers and application decisions, including situations in which the number of available positions is limited. The main domain entities are **Offer**, **Student**, and **Application**.

b. Purpose of the System

The main purpose of InternHub is to simplify the internship application process for students while providing a structured way to manage internship offers and application decisions.
Additionally, the system includes an AI-assisted feature that helps students evaluate how well their CV matches the requirements of a specific internship offer. The AI module identifies skills already demonstrated in the CV, missing skills, and possible improvement suggestions.
The AI component is intended only to provide feedback to the student. It does **not** rank, accept, or reject candidates, and final application decisions remain human-controlled.

c. Main Objectives

The main objectives of the system are:

- to provide students with access to internship offers;
- to allow students to submit and manage internship applications;
- to allow internship offers and their available positions to be managed consistently;
- to prevent invalid applications and conflicting decisions through clearly defined business rules;
- to provide AI-based CV feedback without allowing AI to make recruitment decisions;
- to maintain the privacy of CV data processed by the AI component.

2. Stakeholders and Actors

a. Stakeholders

The main stakeholders of the InternHub system are:

➔ **Students**

Students are the primary beneficiaries of the platform. They use InternHub to discover internship opportunities, apply for positions, manage their applications, track recruitment statuses, and receive AI-assisted CV feedback before applying. The user stories explicitly cover CV feedback, browsing offers, withdrawing applications, and tracking application status.

➔ **Recruiters**

Recruiters represent the hiring side of the platform. They are responsible for publishing internship offers, defining the number of available slots and deadlines, evaluating candidates, accepting or rejecting applications, and manually closing offers when necessary.

➔ **Companies/Employers**

Companies are indirect stakeholders represented in the system through recruiters. They benefit from being able to publish internship opportunities and manage the recruitment process in a structured way.

b. Primary Actors

Student

The Student is a primary actor of InternHub.
A student can:

- browse active internship offers;
- view internship opportunities;
- apply for an internship;
- withdraw an application when permitted;
- view and track all submitted applications;
- receive AI-generated feedback on their CV for a specific internship offer.

The system must also enforce application-related restrictions, such as allowing only one active application per offer, limiting the number of active applications, and rejecting applications submitted after an offer deadline.

Recruiter

The **Recruiter** is the second primary actor of InternHub.
A recruiter can:

- publish internship offers;
- define the offer title, company, requirements, deadline, and number of available slots;
- evaluate applications;
- accept or reject candidates;

- manage situations where all available slots have been filled;
- manually close an internship offer before its deadline.

The recruiter is also involved in concurrency-sensitive operations. For example, if two recruiters attempt to accept different candidates for the final available slot at the same time, only one candidate may be accepted while the other must be placed on the waiting list.

c. Supporting System Actor

Local AI Service (Ollama)

The Local AI Service is a supporting technical actor used by InternHub for CV analysis.
It receives:

- the student's CV text;
- the requirements of a selected internship offer.

It returns structured feedback containing:

- **matchedSkills**
- **missingSkills**
- **suggestions**

d. Actor Summary

Actor	Type	Main Responsibilities
Student	Primary human actor	Browse offers, apply, withdraw applications, track statuses, request CV feedback
Recruiter	Primary human actor	Publish offers, evaluate applications, accept/reject candidates, close offers
Local AI Service	Supporting system actor	Analyze CVs and generate advisory feedback

4. Functional Requirements

The functional requirements describe the main actions that the InternHub system must support for its users. They are derived from the project brief and the defined user stories.

FR - 01 : Student Profile Creation

The system shall allow a student to create a student profile that can be used to interact with internship offers and applications. The project brief explicitly includes `CreateStudent` as one of the main business slices.

FR - 02 : Internship Offer Publication

The system shall allow a recruiter to publish an internship offer. An internship offer must include at least:

- title;
- company name;
- requirements;
- deadline;
- maximum number of available slots.

The offer must also contain a Version property used for optimistic concurrency control.

FR - 03 : Browse Internship Offers

The system shall allow students to browse active internship offers in order to discover suitable internship opportunities.

FR - 04 : View Internship Offer Details

The system shall allow a student to view the details of a specific internship offer. The detailed view should expose the information required for the student to understand the opportunity and decide whether to apply.

FR - 05 : Submit an Application

The system shall allow a student to apply for an active internship offer. Before accepting the application, the system must verify that:

- the offer deadline has not passed;
- the student does not already have an active application for the same offer;
- the student has not exceeded the maximum number of active applications.

The current requirements specify a maximum of five active applications per student and one active application per offer.

FR - 06 : Withdraw an Application

The system shall allow a student to withdraw an existing application when withdrawal is permitted by the application status. Withdrawal is allowed when the application is in the Submitted or Shortlisted state and is not allowed after the application has been Accepted or Rejected. If a student withdraws a Shortlisted application, the system shall notify the recruiter so that another candidate may be considered.

FR - 07 : Evaluate Applications

The system shall allow a recruiter to evaluate an application and change its status to Accepted or Rejected. If no internship slots are available when the recruiter attempts to accept a candidate, the system shall assign the Waitlisted status instead of Accepted.

FR - 08 : View Student Applications

The system shall allow a student to view all of their internship applications. For each application, the system shall display:

- company name;
- offer title;
- current application status.

The possible statuses currently defined are: Submitted, Shortlisted, Accepted, Rejected, Waitlisted, and Withdrawn. The applications shall be ordered by application date in descending order.

FR - 09 : Close an Internship Offer

The system shall allow a recruiter to manually close an internship offer before its original deadline. Once an offer is closed:

- no new applications may be submitted;
- existing applications remain active;
- existing applications may continue through the recruitment process.

FR - 10 : AI-Based CV Feedback

The system shall allow a student to request automated CV feedback for a specific internship offer. The system shall analyze:

- the student's CV text;
- the requirements of the selected internship offer.

The result must contain the following structured fields:

- matchedSkills;
- missingSkills;
- suggestions.

Each missing skill must be associated with an actionable recommendation describing how the student could acquire or better demonstrate that skill. The AI functionality is advisory only and must not automatically rank or reject candidates.

4. Business Rules

Business rules define the conditions and constraints that govern how internship offers and applications behave within InternHub.
They must be enforced independently of the user interface.

BR-01 — One Active Application per Offer

A student may have at most one active application for the same internship offer at any given time.
The system must reject any attempt to create another active application for the same student and the same offer.

BR-02 — Maximum Number of Active Applications

A student may have at most five active applications across the entire system at any given time.
If the student has already reached this limit, any new application attempt must be rejected until one of the active applications is no longer considered active.

BR-03 — Application Deadline

A student may apply for an internship offer only before its deadline.
If the deadline has passed, the system must reject the application.

BR-04 — Closed Offers Cannot Receive New Applications

A recruiter may manually close an internship offer before its original deadline. Once an offer is closed:

- no new applications may be submitted;
- the original deadline is no longer relevant for accepting new applications;
- existing applications remain valid;
- existing applications may continue through the recruitment process.

BR-05 — Application Withdrawal

A student may withdraw an application only when its current status is:

- Submitted;
- Shortlisted.

Withdrawal is not permitted once the application has been accepted or rejected.

BR-06 — Withdrawal of a Shortlisted Application
If a student withdraws an application whose status is Shortlisted, the system must notify the recruiter. The recruiter may then decide whether another candidate should be promoted to the shortlist.

BR-07 — Application Statuses
An application may have one of the following statuses:

- Submitted;
- Shortlisted;
- Accepted;
- Rejected;
- Waitlisted;
- Withdrawn.

The system must ensure that every application has a valid status from this set.

BR-08 — Recruiter Decisions
A recruiter may evaluate an application and decide to:

- accept the candidate;
- reject the candidate.

The resulting application status must reflect the recruiter’s decision, subject to the available-slot rules.

BR-09 — Limited Internship Slots
The number of accepted candidates must not exceed the maximum number of available slots defined for the internship offer. If a recruiter attempts to accept a candidate when no slots are available, the candidate must receive the Waitlisted status instead of Accepted.

BR-10 — Concurrent Acceptance of the Last Available Slot
If exactly one internship slot remains and two recruiters attempt to accept two different candidates at the same time, the system must prevent both candidates from occupying the same final slot.
Optimistic concurrency must be handled using the offer's Version property. As a result:

- one candidate is accepted;
- the other candidate is placed on the waiting list.

BR-11 — Existing Applications After Offer Closure
Closing an internship offer must not cancel or invalidate applications that were submitted before the offer was closed. Existing applications may still transition through the recruitment process, for example from Submitted to Shortlisted, Accepted, or Rejected.

Rule Requiring Clarification

There is one point that should be clarified before implementation. The original project brief states that when a withdrawal frees a slot, the first candidate in the waiting list is promoted. However, the current user stories state that withdrawal is allowed only for Submitted or Shortlisted applications and specifically disallow withdrawal once the candidate has been accepted. Therefore, the team should decide whether:

- an Accepted candidate should also be allowed to withdraw, allowing automatic promotion from the waitlist;
- or accepted candidates cannot withdraw, in which case the automatic waitlist promotion rule needs to be reconsidered.

5. Non-Functional Requirements

The following non-functional requirements define quality attributes and technical constraints that affect how InternHub must operate. They are based on the project brief and the current user stories; no performance target is added where the source material does not specify one.

NFR-01 — CV Data Privacy

The system shall not store the CV text sent to the AI component in the application database.
The system shall not write the CV text to application logs.

NFR-02 — Local AI Processing

CV analysis shall be performed through a local Ollama model so that the CV text does not leave the local execution environment.

NFR-03 — Concurrency and Data Consistency

The system shall preserve the configured internship slot limit when multiple application decisions are processed concurrently.
Optimistic concurrency shall be used through the offer Version property so that two simultaneous acceptance operations cannot consume the same final slot.

NFR-04 — Deterministic Automated Testing

Unit and integration tests shall not depend on a live AI model or on Ollama being available.
A deterministic fake implementation shall be available for AI-related tests so that test results are repeatable.

NFR-05 — Expected Error Handling

- Expected business errors shall be represented using Result.

6. AI Requirements

The AI requirements define the functional and behavioral expectations for the AI-based CV feedback component in InternHub.

AIR-01 — Advisory AI Nature

The AI component shall act strictly as an advisory tool for students. It shall not automatically make recruitment decisions, rank candidates, accept candidates, or reject applications.

AIR-02 — Input Data for CV Analysis

The AI analysis module shall accept as input:

- the student's CV text;
- the requirements string of the selected internship offer.

AIR-03 — Structured AI Output Format

The AI component shall return feedback structured strictly into three parts: matchedSkills, missingSkills, and suggestions.

AIR-04 — Actionable Recommendations for Missing Skills

For each identified missing skill, the AI module shall provide at least one actionable recommendation advising the student on how to acquire or demonstrate that skill.

AIR-05 — Local Model Execution

AI requests shall be processed using a locally hosted Ollama model to ensure that no CV data leaves the internal system environment.

AIR-06 — CV Persistence Prohibition

The system shall not persist raw CV text provided for AI analysis in the database.

AIR-07 — Logging Exclusion for CV Text

The system shall prevent raw CV content from being written to application logs or audit trails.

AIR-08 — Test Isolation from Live AI Model

Automated unit and integration test suites shall not depend on a running Ollama instance or external AI availability.

AIR-09 — Deterministic AI Test Doubles

AI integration tests shall use deterministic fake/mock implementations to guarantee predictable, repeatable results.

AIR-10 — Graceful Handling of AI Service Failure

If the local AI service is unavailable or fails to generate a response, the system shall return an explicit error result without affecting standard application workflows.

AIR-11 — Language Support for CV Analysis

The AI feedback module shall support analyzing CV text and offer requirements in English and Romanian.

AIR-12 — Independent Feedback Generation

Generating CV feedback shall be an independent on-demand action and shall not require submitting an application.

6.1 Requirement Requiring Clarification

It needs to be clarified whether AI feedback results should be cached temporarily during a user session or re-generated on every single student request.

7. Data and Domain Requirements

The domain model defines the key entities, their attributes, and their operational constraints within the InternHub system.

Offer Entity

An Offer represents an internship opportunity published by a recruiter. It contains:

- Id (unique identifier);
- Title and Company Name;
- Requirements description;
- Deadline and Maximum Available Slots;
- IsClosed status flag;
- Version property for optimistic concurrency control.

Student Entity

A Student represents a candidate user on the platform. It contains profile details and tracks submitted applications.

Application Entity

An Application links a Student to an Offer. It contains:

- Id, StudentId, and OfferId;
- Application Date;
- Status (Submitted, Shortlisted, Accepted, Rejected, Waitlisted, Withdrawn).

8. System Constraints

The system development and operation are subject to the following technical and architectural constraints:

- **Local Execution Context:** AI processing must run strictly using a local Ollama instance without external cloud API dependencies to ensure privacy.
- **Data Retention Policy:** CV texts provided for AI feedback must not be persisted in the main database or written into execution logs.
- **Concurrency Control:** Concurrency management for slot limits must rely on optimistic concurrency via offer versioning.

9. Assumptions and Open Questions

Assumptions

- Students submit CV content in plain text format when requesting AI feedback.
- Recruiters are trusted human actors responsible for manual candidate decisions.

Open Questions

- Alignment between application withdrawal rules for Accepted candidates and automatic promotion from the waitlist needs final stakeholder confirmation.

10. Requirement Prioritization

Requirements are categorized by priority according to core business impact:

- **High Priority (Core MVP):** Offer publication, application submission, application limits (BR-01, BR-02), application evaluation, and concurrency control for slots.
- **Medium Priority:** Application withdrawal, offer manual closure, and AI-assisted CV feedback.
- **Low Priority:** Advanced AI suggestions and enhanced analytical reporting.

11. Traceability and Validation

To ensure quality and compliance with specified requirements:

- **Unit & Integration Tests:** Core domain business rules (BR-01 to BR-11) will be validated using automated unit tests with deterministic test doubles for the AI service.
- **Concurrency Validation:** Parallel execution tests will verify that optimistic locking on the offer Version prevents overbooking available slots.
- **Privacy Audit:** Automated check routines will ensure log files and application stores contain no raw CV payload data.